

QUIZZICLES-2nd Edition

July 2025
Question Paper

RULES:

- The exam duration is 24 hours
- Start Time: 25/07/2025, 6:00pm IST
- This is an open-book and open-internet exam.
- The total number of questions is 43.
- The **first 10 questions** have a maximum of 2 attempts each.
- The remaining 33 questions allow only 1 attempt.
- All answers are integers between 0 and 9999 (inclusive).
- Answers must be submitted on the **Quizzicles Exam Portal**.
- **Marking Scheme:** The exam consists of 43 questions carrying a total of **200 points**. The marks per question are as follows:

Q1	1 mark	Q16	2 marks	Q31	6 marks
Q2	3 marks	Q17	4 marks	Q32	5 marks
Q3	2 marks	Q18	5 marks	Q33	6 marks
Q4	4 marks	Q19	3 marks	Q34	8 marks
Q5	4 marks	Q20	5 marks	Q35	5 marks
Q6	2 marks	Q21	3 marks	Q36	7 marks
Q7	2 marks	Q22	4 marks	Q37	6 marks
Q8	2 marks	Q23	5 marks	Q38	10 marks
Q9	1 mark	Q24	5 marks	Q39	6 marks
Q10	3 marks	Q25	4 marks	Q40	10 marks
Q11	2 marks	Q26	3 marks	Q41	10 marks
Q12	3 marks	Q27	6 marks	Q42	9 marks
Q13	2 marks	Q28	5 marks	Q43	7 marks
Q14	4 marks	Q29	9 marks		
Q15	2 marks	Q30	5 marks		

§1 What's the time, sorry, angle?

Between 12:00 PM and 1:00 PM, how many times are the minute and hour hands of a clock at exactly 120° apart?

§2 Let's play a game

Getting down the rabbit hole, you finally reach the Wonderland. After a long search, you finally find Alice. As usual, Alice and Bob are playing games (and devising the winning strategies). In order to test your skills, Alice and Bob decide to team up against you in a game of modified Nim. In this game there are n piles of stones (n is an even natural number). A player in their turn needs to choose exactly $n/2$ nonempty piles and independently remove a positive number of stones from each of the chosen piles. They can remove a different number of stones from each of the chosen piles in a single turn. The first player unable to make a move loses (when there are less than $n/2$ nonempty piles). Alice and Bob move first. Since it is your first game against them, they allow you to choose the number of stones to be kept initially in the n^{th} pile, but out of their discretion, you can only keep integral multiples of 3 stones in this pile. The remaining $n - 1$ piles have $[30, 32, 35, 30, 32, 50, 30, 35, 30, 32, 30, 30, 30]$ stones in order. (They choose n to be 14) What is the minimum number of stones you would place in the last pile so that you can win? If you think it is impossible for you to win, report your answer to be 0.

§3 Power of Observation

Two concentric circles with a common center O have radii r_1 and r_2 , where $r_1 < r_2$.

Let A and D be **variable points** on the **outer circle** (radius r_2). Let B and C be **variable points** on the **inner circle** (radius r_1).

The four points A, B, C, D lie on the same straight line in the order A, B, C, D . A **variable point** P is chosen on the outer circle such that $\angle BPC = \alpha$.

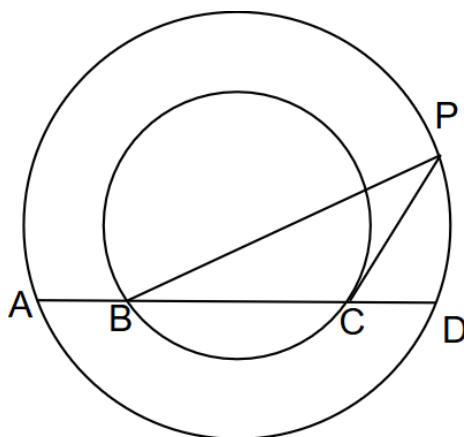


Figure 1: A, B, C, D collinear; B, C on inner, A, D on outer

Given Values:

- $r_1 = 1$
- $r_2 = \sqrt{3}$

Cases:

1. $\alpha = 40^\circ$
2. $\alpha = 60^\circ$
3. $\alpha = 75^\circ$

Task: For each of the three cases above, determine A_1, A_2 , and A_3 , which represent the lengths of the chord AD in each respective case, using the rules provided.

Important Instructions

In Case i , the chord length A_i is determined as follows:

1. If **multiple values** of AD are possible (i.e., length is not uniquely determined), report: $\boxed{A_i = 10}$
2. If **no valid chord** AD is possible for the given angle, report: $\boxed{A_i = 6}$
3. If a **unique value** of AD is determined, report its exact magnitude as A_i .

Final Submission Instructions:

You must submit **only the final boxed product** of the three answers — **nothing else**. *Do not submit the individual values of A_i for each case.*

- Use the approximation: $\sqrt{3} = 1.732$
- After calculating the product $A_1 \times A_2 \times A_3$, **report your final answer as the floor of this product**

$$\boxed{[A_1 \times A_2 \times A_3] = ?}$$

§4 A Tale of Two Ds: Midpoint or Bisector on BC

In triangle ABC , point D lies on side BC .

1. Suppose D is the midpoint of BC . Given that $AB = 6$ cm, $AC = 10$ cm, and $BC = 8$ cm, find the length of segment AD and store it in x .
2. Now suppose AD is the angle bisector of $\angle BAC$. Using the same triangle, again find the length of segment AD and store it in y .

Compute the **square of the product**, that is, compute $(xy)^2$ and report the answer **without units**.

§5 FIFA 2022

It is 2022 FIFA world cup penalty shoot-outs, Messi has to score a goal against French goalkeeper, Hugo Lloris (height 6 ft 2 in). Lloris can dive in any direction at a maximum speed of 3.5 m/s and has a reaction time of 0.1 s. Consider the goal post to be infinitely large and the shootout point to be 12 yards away from the goal post. Also, assume gravity to act only on the ball and not on the goalkeeper. Let the minimum speed (in m/s) at which Messi must kick the ball to score a goal be x , considering the ball to be a point-sized object and Lloris to be a stick-sized object. Find $[x]$ (where $[.]$ is the Greatest Integer Function).

§6 Iron Dome

An air defense system can fire at incoming threats with missiles having velocities, which can reach up to mach 20. Imagine the missiles once launched, accelerating under the influence of gravity alone. What is the volume of the region it can protect? If the volume is $a \times 10^4 km^3$, report the greatest integer less or equal to a .

Given: Speed of sound = 343 m/s

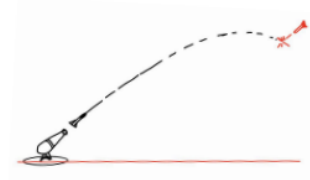


Figure 2: Iron Dome

§7 Maximize the minimum value

Let x, y be positive reals. Find the maximum possible value of $g(x, y)$ where

$$g(x, y) = \min \left\{ y + \frac{120}{x}, \frac{1}{y}, x \right\}$$

§8 Remainders Puzzle

What is the smallest positive integer that leaves a remainder of:

1 when divided by 2

2 when divided by 3

3 when divided by 4

...

9 when divided by 10

§9 A Tale of Two Cities

A city of 10,000 people has 50% male and 50% female population

Average male life expectancy is 72 years

Average female life expectancy is 78 years

One day, the city merges with a nearby town of 10,000 people with 60% male and 40% female population.

Male expectancy is 74

Female expectancy is 80

After the merge, the overall average life expectancy changes.

Q: What is the new average life expectancy of the combined population?

If the value is x years, report 10x as the answer.

§10 Just a Number

A number has 4 digits. It satisfies the following conditions: All digits are nonzero and distinct. The number is divisible by each of its digits. Reversing the digits gives a new number that is also divisible by each of its digits. The sum of the digits is a multiple of 9. What is the smallest such number?

§11 Pull...Pull...Pulley

An infinite series of pulleys and masses is arranged such that:

$$M = t, \quad m_i = t^{i-1} \quad \text{for } i > 0, \quad \text{with } 0 < t < 1.$$

Now, we attach two more pulleys to the mass M . These pulleys support masses:

$$m_{-j} = r^j \quad \text{for } j = 1, 2, 3,$$

with $r > 1$.

Given: $t = \frac{1}{5}$, $r = 2$, $g = 9.81 \text{ m/s}^2$

Question: Find the magnitude of acceleration of mass m_{-3} , i.e., $|a_{-3}|$, and return:

$$\lfloor 100 \cdot |a_{-3}| \rfloor$$

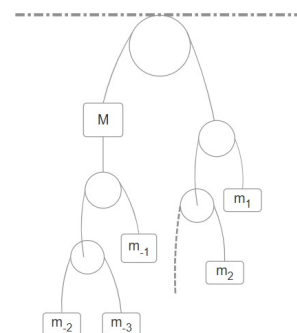


Figure 3: The pulley system

§12 School time!

Bibu is a smart guy. To prove his smartness, Bibu challenged his classmates with an interesting problem. He gave 2 numbers: **num1** = 18, **num2** = 24. Then he gave a long list of numbers

$$C = [59, 9, 117, 14, 29, 11, 40, 98, 604, 45].$$

Now Bibu says his friends can add **num1** and/or **num2** any number of times to any of the elements of the list C . Each such addition is considered as 1 operation. Then he asks to find what can be the minimum possible difference between the maximum and minimum elements of the array after the operations. Can you tell the answer?

§13 Painting Job

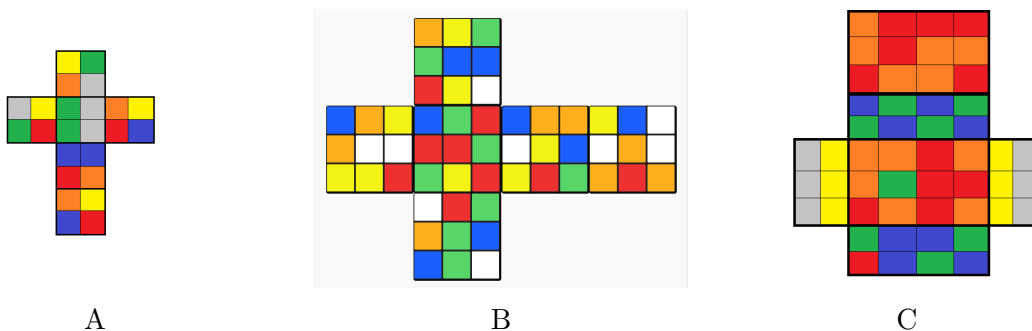
You are working in a table manufacturing company and your job is to paint the tables. To simplify your job, the company has made a machine that can paint a circular area of radius 1 meter in one operation. You have to paint a square table of dimension 10 m x 10 m. To reduce the cost and time, you have to paint it in the minimum possible operations. Find the minimum number of operations needed.

§14 Tintin: The Adventure of the Cuboids

It seems Captain Haddock cannot get over his obsession for Rubik's Cube. So, Professor Calculus creates cubes/cuboids of different dimensions for him to solve. Having come to visit Tintin at Marlinspike hall, Thomson and Thompson try to test their skills at solving these cubes/cuboids. Failing miserably, they ask Tintin for help. Tintin, using his aptitude skill at first tries to know, "Are these cubes solvable?" Help Tintin find the answer to his query. The cubes/cuboids are of dimensions A) $2 \times 2 \times 2$, B) $3 \times 3 \times 3$ and C) $2 \times 3 \times 4$. If a cube is solvable report 6, else report 5. Finally report the answer as the three digit number representing the solvability of A, B and C respectively.

Note: A cube is solvable if, by rotating its faces, it is possible to reach a configuration where each face consists entirely of cubies of a single colour.

The net diagram of the cubes/cuboids are given below.



§15 Cycles and permutations

Find number of functions f from the set $\{1, 2, \dots, 8\}$ such that $f^3(x) = x$.

§16 Scanning-Blade Calibration

Mr. Shaan at Quizzicles Engineering has developed a revolutionary **Scanning-Blade** device, projecting two beam envelopes on a flat metallic surface:

- A **dual-blade beam**:

$$f(a, b, r) : (x - a)^2 + (y - b)^2 = (x + y - r)^2,$$

with tunable real parameters (a, b, r) .

- A **safety-circle beam**:

$$g(r) : x^2 + y^2 = r^2.$$

Engineers have discovered that for certain parameter choices, the dual-blade beam $f(a, b, r)$ splits into two real straight lines. Let

$$S = \left\{ (a, b, r) \in \mathbb{R}^3 : f(a, b, r) \text{ splits into two real lines} \right\}.$$

Now considering only those parameters, define the **safe-zone area** $A(a, b)$ as the area lying **in the first quadrant** (i.e., where $x \geq 0, y \geq 0$), bounded **between**:

- the **pair of lines** defined by $f(a, b, r)$, and
- the **circular arc** of $g(r)$,

for all $(a, b, r) \in S$.

That is,

$$A(a, b) = \iint_{\substack{\text{region between the lines of } f(a, b, r) \\ \text{and the circle } x^2 + y^2 = r^2 \\ \text{within the first quadrant: } x \geq 0, y \geq 0}} dA.$$

To calibrate the system, Quizzicles Engineering evaluates:

$$T = \sum_{a=1}^{10} \lfloor A(a, 2a) \rfloor,$$

where for each a , the corresponding value of r is chosen so that $(a, 2a, r) \in S$.

Question:

Compute the integer calibration sum:

$$T = \sum_{a=1}^{10} \lfloor A(a, 2a) \rfloor.$$

Note: The area is defined only within the first quadrant, and bounded by the curves $f(a, b, r)$ and $g(r)$. Determining when f becomes a pair of lines is part of the problem.

§17 The Spiral Web

Some spiders spin spiral orb webs, with radial lines to support the spiral.

Mathweb is one such spider who loves making spiral webs and has a curious mind. He decides to spin a special spiral. He lies at the centre of an 8×8 square and has woven numerous radial lines from the centre to the sides of the square. Now he starts weaving a spiral web such that any two consecutive points on a radial line are exactly 1 unit apart (except at the centre). This condition is even valid for the centre in the case of the radial line along his initial direction of motion.



Figure 5: A spiral orb web

A rough schematic showing Mathweb's idea is given below:

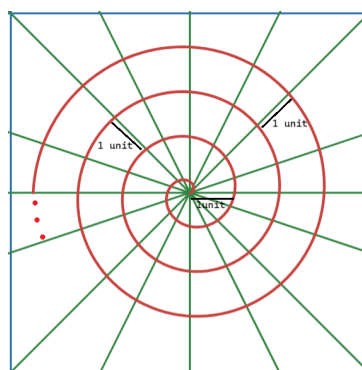


Figure 6: The spiral

Let A be the point where the spiral intersects the square and Mathweb stops spinning (grazing the side is fine). Let x be the distance of the point A from the nearest vertex of the square. Find the value of $\lfloor 100 \cdot x \rfloor$ (Greatest integer function).

NOTE: You may need to use numerical methods

§18 Find the Number !

Let N be the largest 10-digit number such that:

- All its digits are distinct (i.e., every digit from 0 to 9 is used exactly once),
- N is divisible by 11.

Compute the value of $N \bmod 10,000$, i.e., the remainder when N is divided by 10,000.

§19 Havoc of Operations

There is a number represented by 100 digits, all digits initially being 0. 100 people perform the following operation one by one: The i 'th person replaces all 1's by 0's and vice versa, at the positions divisible by i . After all the operations, find the number of

digits with value 1. Note : Position is counted from the rightmost digit, the rightmost digit being position 1.

§20 All numbers representable

Let S be the sum of all positive integers ≤ 2025 which can be represented in the form of $x^2 + y^2 + z^2 + z$ for integers x, y, z . Find the remainder when S is divided by 10000

§21 Electron Lens

Lensing in physics means to bend a wave or particle trajectory by some field or changing medium. While you may have seen (or are even seeing through) lenses that focus light, a typical "optical" lens isn't the only kind of lens to exist. There are phenomena as gravitational lensing, magnetic lensing, electrostatic lensing, acoustic lensing and so on. In this question, we shall talk about electrostatic lensing.

An electrostatic lens is a device that focusses charged particles. For our case, we shall focus on focussing electrons. In this problem, we construct a simple electrostatic lens.

Consider a metallic ring of radius $R = 20\text{cm}$ and width $d = 5\mu\text{m}$ as shown in the figure:

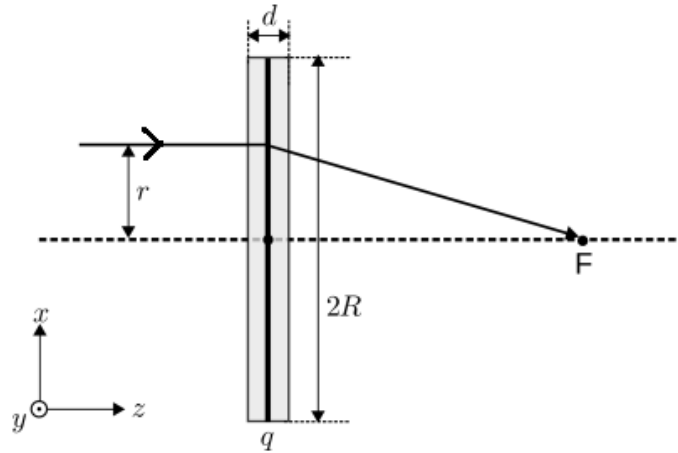


Figure 7: The electrostatic lens

Electrons are coming from a very large distance parallel to the axis of the ring with kinetic energy $E = 2\text{eV}$ at regular intervals. The ring has a constant charge $q = 10\mu\text{C}$ only when electrons are within the width of the ring (i.e., the electrons are within $d/2$ ($d \ll R$) distance from the central plane of the ring, other times it has zero charge. The intervals are such that any electron is acted upon by a force only when it is in this width. Due to the electric field of the ring, electrons bend towards the axis. Consider the paraxial electrons ($r \ll R$). For such electrons, what will be the focal length of this "electrostatic lens" in cm (Answer to nearest integer) ?

Note: The potential of a point at distance r ($r \ll R$) in the plane of a ring of charge q is approximately, $\Phi(r) = \frac{q}{4\pi\epsilon_0 R} \left(1 + \frac{r^2}{4R^2}\right)$.

Given, $\epsilon_0 = 8.854 \times 10^{-12} \text{F/m}$ and $e = 1.602 \times 10^{-19} \text{C}$

§22 Rotating coins

You have 2 identical circular coins (radius = R). One coin is fixed, and the other rolls around its edge externally without slipping. How many full rotations does the moving coin make before returning to its starting point? (Note: The coin rotates around the other without flipping.)

§23 Unfair Die!

A die is rolled 3 times. However, this die is not fair: Faces 1, 2, 3 occur with equal probability. Faces 4, 5, 6 occur with equal probability, but twice as likely as the others. You are told the product of the three rolls is 216. What is the probability that at least one of the rolls was a 6? If the answer is p/q in simplest form, report $p+q$

§24 CONTEST in a CONTEST

In a contest there are 10 questions. A man attempts the paper and achieves a pattern. He does not score less on a question he attempts earlier when compared to a question he attempts later. The marking scheme is as follows -

Case i) completely correct = 10 points.

Case ii) partially correct = $10 - 9$ points (Natural numbers only).

Case iii) unattempted = 0 points.

Case iv) Wrong attempt = -1 point.

A set μ is constructed with each element as follows:- $S = \{\text{Marks in Q1, Marks in Q2, ..., Marks in question Q10}\}$.

Let $100 * a$ be the number of possible elements in μ . What is the value of a rounded off to the nearest integer?

§25 Trek to Mt. Kilimanjaro

You're hiking Mt. Kilimanjaro with 7 checkpoints. At each checkpoint, you either:

Gain elevation equal to a square number (e.g., +1, +4, +9, ...)

Or lose elevation equal to a square number. Your total gain/loss across all checkpoints must equal 0, and no checkpoint can have zero change.

Q: What is the minimum possible total absolute elevation change over the 7 checkpoints?

§26 Detective Time!

Five suspects: A, B, C, D, and E are questioned.

Each says one statement:

A: "B did it."

B: "C didn't do it."

C: "D did it."

D: “E is lying.”

E: “B is lying.”

Only one person is telling the truth.

Q: Who is the murderer? (Answer with 1 for A, 2 for B, ..., 5 for E)

§27 The Whispering Soundboard

A concert grand piano string produces a note with a frequency of $f = \frac{100}{\pi}$ Hz. The string is coupled to a soundboard which behaves like a damped harmonic oscillator.

- Mass ($m = 2.0$ kg),
- Quality factor ($Q = 100$).

The energy stored in the vibrating string is $E = 0.2$ J, and all of it is transferred to the soundboard over one damping time (the time it takes for the energy to reduce by a factor of e). **Calculate the amplitude A (in mm) of oscillation of the soundboard at resonance.** Give your final answer as the nearest **integer** in millimeters.



Concert Piano

§28 Win the Game!

Chintu & Mintu were 2 friends. Being bored, they started to play a game. They have created a list of values and have prepared a set of rules.

Game Rules

- Chintu will start first. Each of the 2 takes values turn by turn.
- If money taken by Chintu is even then he adds it to his score. If odd, then he can't add it to his score.
- If money taken by Mintu is odd then he adds it to his score. If even, then he can't add it to his score.
- Any value once taken by any player, can't be taken by other player later.
- The values can be taken in any order (Not necessary to pick 1st value at beginning or last value at the end).

The player with highest score wins.

If both of them are really smart and thus plays optimally, then who will win?

The list of values is $C = [15, 61, 100, 72, 10, 51, 76, 89, 96, 43, 91]$

Write 0 if you think Chintu wins, 1 if Mintu wins or if it will be draw, write 2 and if it can't be decided print 3.

§29 Can you save their friendship?

Sagnik wants to surprise his friend Ayushi by creating a beautiful hexagonal mosaic art piece for her birthday. He starts with a large regular hexagon board of side length 10 units and a collection of special diamond-shaped tiles.

Each diamond tile is a rhombus with side length 1 unit and interior angles of 60° and 120° , giving each diamond an area of $\sqrt{3}/2$ square units. Sagnik can perfectly tile the entire hexagonal board using these diamond tiles in various arrangements.

During the art creation process, Sagnik discovers an interesting property: whenever three diamond tiles are positioned so that they are pairwise adjacent and form a small regular hexagon of side length 1 unit, he can perform a special **"rotation move."** This move rotates all three diamonds simultaneously by 60° about the center of the small hexagon they form.

Ayushi mentions that she has seen many different diamond tiling patterns of the same hexagonal board and wonders about the relationship between them. This gives Sagnik an idea for a mathematical challenge.

Problem: Sagnik wants to prove that any diamond tiling pattern of his hexagonal board can be transformed into any other diamond tiling pattern using only rotation moves. Find the smallest positive integer N such that any tiling of the hexagonal board can be transformed into any other tiling in at most N moves.

The hexagonal board has side length 10 units, and each diamond tile has the properties described above. A rotation move can only be performed when three pairwise adjacent diamonds form a regular hexagon of side length 1 unit, and the move rotates all three diamonds by 60° about the center of that small hexagon.

Help Sagnik determine **the minimum number of moves needed to transform between any two diamond tilings of his hexagonal art board**, so that he can impress her.

§30 Frixel turns a leaf!

Scientist Frixel is admiring the fragrance of the flowers of *Nyctanthes arbor-tristis*, commonly known as Night-Blooming Jasmine (*Harsinghar/Shiuli*) when he touches the leaves of the plant by mistake. Mesmerized the more, he is! Moving his hand over to the tip of the leaf was easy, but going the opposite way offered more resistance: "The coefficient of friction must be dependent on the velocity (vector)", Frixel reasoned.

Curious, he performed experiments and found that the coefficient of friction, μ depended on the unit velocity vector, $\hat{v}$ as $\mu = \alpha(1 - \hat{v} \cdot \hat{j})$, where $\hat{j}$ denotes the direction towards the tip of the leaf.

Frixel now wants us to do some work! He has given us an



Figure 8: Leaf of *Nyctanthes arbor-tristis*

inclined plane with angle of inclination, $\theta = 45^\circ$. On the incline surface, he placed a very large leaf of *Nyctanthes arbor-tristis* cut into a square. The tip of leaf direction ($\hat{\mathbf{j}}$) is up along the plane. Also, he has measured the value of $\alpha = 1$. Along this inclined surface, he launches a small object with initial speed $u = 20\text{m/s}$ at an angle of $\phi_0 = 30^\circ$ with the $\hat{\mathbf{j}}$ direction. He has given us the challenge to find the speed v of the object when it makes an angle of $\phi = 60^\circ$ with $\hat{\mathbf{j}}$. Report the value of $[10v]$ ($[.]$ denotes the Greatest Integer Function).

§31 Floor is Lava!

Story:

After binge-watching *Squid Game* Season 3, Shaan starts fantasizing about participating in a thrilling, high-stakes game. One evening, he stumbles across a mysterious website offering an enormous cash prize to whoever can win their “fun” game. Shaan, seeking adventure, eagerly signs up.

On the scheduled date, he arrives at the hidden location. The guards usher him to a long staircase labeled from levels 0 to 10, with a boiling, sinister liquid bubbling just below level 0.

Shaan asks, “What’s the game?” One of the guards responds coldly:

“You have to get to the top—level 10.”

Seems easy... until the guard adds:

“But...”

Shaan straightens up.

“You have a biased coin with heads probability p . At each step, you toss the coin. If it’s heads, you go up by one level. If it’s tails, you go down by one.”

Shaan, now a bit uneasy, asks, “Wait... if I start at 0 and get tails on the first toss, I fall into that lava?”

“Yes,” says the guard grimly. “But there’s more. Every two steps you take—regardless of direction—the lava rises by one level. So, if after 2 steps you’re at level 0, you’re eliminated, since the lava has now risen to that level.”

Then comes the twist. The second guard smirks:

“You get to choose the probability p of the coin. But your reward, if you make it to the top, is only

$$1000 \ln\left(\frac{1}{p}\right)$$

units of money.”

Now Shaan must choose a value of $p \in (0, 1]$ that maximizes his expected reward.

Question. Enter the value of $[1000p]$.

§32 Bridge at Night

Four people need to cross a bridge at night. They have one flashlight, and at most two can cross at a time. No one can cross the bridge without one having the flashlight. Time taken for each to cross the bridge are 1, 2, 5, and 10 minutes respectively. When two cross, they move at the slower person's pace. What is the minimum time needed (in minutes) for all four to cross?

§33 Minimum value of floors

What is the least value that the expression

$$\left\lfloor \frac{x_2 + x_3 + x_4}{x_1} \right\rfloor + \left\lfloor \frac{x_1 + x_3 + x_4}{x_2} \right\rfloor + \left\lfloor \frac{x_1 + x_2 + x_4}{x_3} \right\rfloor + \left\lfloor \frac{x_1 + x_2 + x_3}{x_4} \right\rfloor$$

may achieve, where $x_1, \dots, x_4$ are positive real numbers.

§34 What's the missing number

You are given a sequence of 4-digit numbers. Each number in the sequence is formed using only the digits 1 to 9 (no zeros). The numbers follow a hidden transformation rule from one to the next. Here is the sequence:

1. 2351
2. 3642
3. 1426
4. 2863
5. 4135
6. ????

You are to determine the 6th number.

§35 Aniket's Mathematical Challenge

Three brilliant mathematics graduate students—Sagnik, Pranjal, and Srijit—discover a dusty, ancient manuscript hidden deep within the university library's forbidden section. The manuscript bears the signature of **Aniket**, a legendary mathematician who vanished mysteriously decades ago, leaving behind only cryptic clues and unsolved problems.

As they open the manuscript, a ghostly figure of Aniket materializes before them, his ethereal form flickering with mathematical equations. His voice echoes through the library:

"Foolish mortals! You dare to touch my sacred writings? I have spent eternity perfecting the ultimate mathematical secrets—formulas that could revolutionize technology, unlock the mysteries of the universe, and grant unimaginable wealth to whoever possesses them."

*But you three... you are **unworthy** ...because you run after answers instead of enjoying the process ... you love MCQs instead of rigorous proofs!*

The ghost's eyes burn with mathematical fury as he continues:

"However, I shall give you—and any other mathematician brave enough to attempt this challenge—one chance to prove your worth. Hidden within my compendium are prototype formulas for advanced cryptography, quantum computing algorithms, and financial modeling systems that could make their possessor the richest person alive. But first, you must solve this riddle faster than these three graduate students!"

Aniket's ghostly hand waves, and mathematical symbols begin glowing in the air.

The Challenge of Worthiness

The function $f(x)$ is defined for all $x > 0$ and satisfies the following three sacred conditions:

1. $f(x)$ is strictly increasing on $(0, +\infty)$.
2. $f(x) > \frac{-1}{x}$ for $x > 0$.
3. $f(x) \cdot f(f(x) + \frac{1}{x}) = 1$ for all $x > 0$.

Your Task: You must find

$$\lfloor 100 \cdot |f(1)| \rfloor$$

before Sagnik, Pranjal, and Srijit can solve it. Only then will the ghost of Aniket deem you worthy of accessing his compendium of mathematical treasures.

The three graduate students frantically begin their calculations, their pencils scratching against paper as they race against time. Sagnik focuses on the monotonicity condition, Pranjal examines the inequality constraint, and Srijit attacks the functional equation directly.

The ghost of Aniket hovers above, his translucent form crackling with anticipation. *"Solve this, mortal, and claim your destiny. Fail, and join the ranks of the unworthy..."*

Can you determine

$$\lfloor 100 \cdot |f(1)| \rfloor$$

and unlock the secrets that could change your life forever?

Note: This is a race against time. The first to solve correctly will gain access to Aniket's legendary mathematical compendium and the untold riches it contains.

§36 Make a number

Find **sum of the digits** of the largest integer less than 1,000,000 such that:

- It uses each digit from 0 to 9 at most once,
- It does not contain the digit 7,
- It is divisible by 3, 5, and 11

§37 Joy loves ties

Joy has n ties. He will be joining college this summer. But as he belongs from economically weaker section, his bag can only accommodate upto 16 ties in his bag. He has a lot of friends, 6996 to be precise. He sends his list of ties to all his friends and receives an order or ranking of all the ties from each of them. Ranks go from $(1, 2, 3, \dots, n)$. But he himself has to take the final decision. He decides not to upset anybody's ranking list and brings up a method. Let 't' be a tie he doesn't include in his bag. Then in all the rankings from all his friends, there must be at least 1 tie in the bag that had higher preference in at least 3498 friends compared to 't'. You have to calculate the sum of all n in the range $[0, 69420]$ for which he can fill his bag (not necessarily completely) in his own unique method and **report the sum of the digits of this sum** as answer. *NOTE: All ties are distinct.*

§38 The Island of Truth and Lies: The Divisible Hat Conundrum

On a remote island, all inhabitants are either Knights (who always tell the truth) or Knaves (who always lie). You encounter three islanders, A, B, and C, each wearing a hat with a distinct positive integer on it. Each islander can see the numbers on the other two hats, but not their own. They are all perfect logicians, meaning they can deduce all logical consequences of what they know.

You ask each islander, in turn, about their hat number. Their responses are as follows:

- **A says:** "My number is the smallest of the three numbers, and it is a multiple of 4."
- **B says:** "My number is the largest of the three numbers, and it is a prime number."
- **C says:** "My number is the middle number, and it is divisible by 5."

Given that exactly one of the three islanders is a Knave, and the sum of the three hat numbers is 20, what is the product of the numbers on hats A, B, and C?

§39 Guess the numbers

Rohit and Mohit are playing a game in which Mohit chooses two numbers between 1 and 34 (both inclusive), and Rohit has to guess the numbers. The rules are that Rohit has to call out an array of continuous integers (for example $[1, 2, 3, 4, 5]$, $[17, 18, 19, 20, 21]$) and Mohit will either say yes, no, or some corresponding to whether both, none, or one of the numbers he chose lies in the array, respectively. Rohit wins when he gets to know the numbers and does not have to call out another array. Find the minimum number of call-outs that Rohit has to give in order to guarantee his win?

§40 Principal Components Under Pressure

Sagnik was conducting an experiment to estimate **annual medical expense scores** using various health-related features. From a dataset of 100 entries, he selected 5 samples, each having 5 features: **Age**, **Body Mass Index**, **Smoker** (1 stands for Yes while 0 for No), **Exercise Hours per Week**, and **Chronic Conditions**. The sample data is shown below:

Person	Age	BMI	Smoker	Exercise_Hours_per_Week	Chronic_Conditions	Annual_Medical_Expenses (INR)
Alice	52	28.4	1	1	2	54,200
Bob	36	23.1	0	5	0	12,000
Carol	45	31.6	1	0	1	67,500
David	29	26.2	0	4	0	9,000
Ellen	61	27.8	0	2	3	80,000

Table 1: Sample dataset for predicting annual medical expenses (values in INR)

Sagnik's friend **Pranjal**, who had just learned about **Principal Component Analysis (PCA)** and the **Loève Theorem**, challenged him to reduce the dimensionality of the dataset from **5D** to a **3D representation** using the theorem.

Sagnik computed the **covariance matrix** of the dataset (without applying Bessel's correction), using the formula:

$$\mathbf{Cov}(X) = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^\top$$

where $\mathbf{x}_i$ is the i -th sample vector, and μ is the mean vector of all samples.

From this covariance matrix, he obtained **eigenvalues** e_1, e_2, e_3, e_4, e_5 and corresponding **eigenvectors** $\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4, \vec{v}_5$. Using the **top 3 eigenvectors** (corresponding to the **largest eigenvalues**), he formed the **principal components**: PC_1, PC_2, PC_3 , in **decreasing order of the variance** they explain in the data.

Pranjal reminded him that the covariance matrix should ideally use **Bessel's correction**. After applying the correction, Sagnik recalculated and obtained new **eigenvalues** $e'_1, e'_2, e'_3, e'_4, e'_5$ and corresponding **eigenvectors** $\vec{v}'_1, \vec{v}'_2, \vec{v}'_3, \vec{v}'_4, \vec{v}'_5$, forming new **principal components**: PC'_1, PC'_2, PC'_3 .

Help Sagnik, compute the value of Q :

$$Q = 100 \times \left(\sum_{i=1}^3 \left| \frac{e'_i}{e_i} \right| + \sum_{i=1}^3 \|PC'_i - PC_i\| \right)$$

Note:

- $|\cdot|$ denotes the absolute value (modulus) of a scalar.
- $\|\cdot\|$ denotes the Euclidean norm (i.e., magnitude) of a vector.

§41 The Grand Space Exploration!

You have been granted the wish to remain eternally young! After a number of scientific marvels in space travel, you decide to travel to some other star system. You choose a star named “Sola” as your destination (you decide to explore the “Sola” system). “Sola” is located 7.5 lightyears away from the Sun. Prior to your adventure, you study the skymap for the day of launch. The rough sketch of the map is shown here:

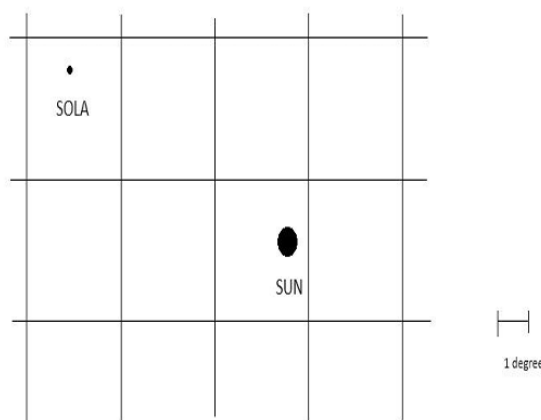


Figure 9: The Sky Map

You need to drive your space shuttle towards ‘Sola’. The initial direction of motion is perpendicular to the ecliptic (plane containing Sun and Earth). Let the velocity of launch of your space shuttle be V . Find $10000(V/V_{\text{esc}} - 1)$ where V_{esc} is the speed that the shuttle needs to just leave the solar system. The mass of Sun, $M_S = 1.989 \times 10^{30} \text{ kg}$, the mass of Earth, $M_E = 5.972 \times 10^{24} \text{ kg}$, the radius of Earth, $R_E = 6.371 \times 10^6 \text{ m}$, the astronomical unit, $a = 1.496 \times 10^{11} \text{ m}$ and the gravitational constant, $G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$. Answer to the nearest integer. *Note: Consider all velocities relative to Sun. Also, we consider the gravitational effects of Sun and Earth only.*

§42 The Enigmatic Sequence

Consider a sequence of positive integers, $a_1, a_2, a_3, \dots$, defined by the following rules:

1. $a_1 = 1$
2. For $n > 1$, a_n is the smallest positive integer such that a_n has not appeared in the sequence $a_1, a_2, \dots, a_{n-1}$, and the sum $a_n + n$ is a perfect square.

What is the value of a_{10} ?

§43 Endurance’s 180° Dive

In Christopher Nolan’s cinematic masterpiece *Interstellar*, the intrepid crew of the *Endurance* embarks on a desperate mission to save humanity. A pivotal and visually stunning sequence involves their harrowing, fuel-efficient maneuver around Gargantua, a rapidly

spinning supermassive black hole. This **gravitational slingshot** allowed them to achieve speeds that would have been impossible with their remaining onboard propulsion, effectively stealing momentum from Gargantua's colossal gravitational well.

Imagine a simplified, idealised version of this maneuver in a two-dimensional plane. The *Endurance*, having expended much of its fuel, relies solely on Gargantua's gravitational assist to reach a distant, crucial wormhole. For the purposes of this problem, consider Gargantua as a point mass moving with a constant velocity $\vec{v}_P$ relative to the distant star system from which the *Endurance* departed. The *Endurance* executes a perfect, frictionless hyperbolic flyby. Its trajectory is engineered such that it approaches Gargantua, swings around it due to gravitational pull of Gargantua, and departs with its velocity vector relative to Gargantua undergoing a 180° change in direction. This specific maneuver is chosen to **yield the maximum possible** increase in the final speed of the *Endurance* relative to the star system.



Gargantua

The Challenge:

The *Endurance* is observed at two points, with velocities measured **relative to the distant star system**:

1. Just before its gravitational interaction with Gargantua begins, traveling with an initial velocity $\vec{v}_{Ei}$ with speed $V_{Ei} = 10 \text{ km/s}$.
2. Immediately after completing its slingshot maneuver around Gargantua, achieving a final velocity $\vec{v}_{Ef}$ with speed $V_{Ef} \text{ km/s}$.

During the maneuver, Gargantua (the black hole) is moving with a constant velocity $\vec{v}_P$ with a speed of $V_P = 25 \text{ km/s}$ relative to the distant star system.

Assuming that the gravitational interaction is perfectly elastic (no energy loss from the *Endurance* during the encounter, only exchange of momentum with Gargantua), determine the **maximum possible final speed** $V_{Ef} = |\vec{v}_{Ef}|$ of the *Endurance* relative to the distant star system.

End of Question Paper
